

Uniform Permanence in Weakly Reversible Mass-Action Systems

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October 5, 2026

Abstract

We prove the permanence conjecture for finite weakly reversible mass-action systems with fixed positive reaction rates. Every positive stoichiometric compatibility class admits one compact convex forward-invariant set that every positive trajectory in that class enters in finite time, even when the class is unbounded. The set depends only on the network, the rates, and the class; the entry time may depend on the initial state. Thus, after its entry time, every concentration satisfies the same positive lower and finite upper bounds throughout that class.

1 Introduction

Weak reversibility requires every reaction in a chemical reaction network to have a directed return path. It is natural to ask whether this graph condition alone prevents concentrations from approaching zero or infinity. Permanence asks for more: after a transient, all trajectories allowed by the same conservation laws must satisfy the same positive lower and finite upper bounds. We prove this conclusion for arbitrary finite weakly reversible networks with fixed positive reaction rates.

1.1 The problem and main result

Let $d \geq 1$. A finite reaction network consists of a set of *complexes* $C \subset \mathbb{Z}_{\geq 0}^d$ and a set of directed *reactions* $R \subset C \times C$. We write $y \rightarrow y'$ for $(y, y') \in R$. The network is *weakly reversible* if for every $y \rightarrow y' \in R$ there is a directed path from y' to y . A *linkage class* is a connected component of the underlying undirected graph. Assign a positive constant $k_{y \rightarrow y'}$ to each reaction. The mass-action equation is

$$\dot{x} = f(x) := \sum_{y \rightarrow y' \in R} k_{y \rightarrow y'} x^y (y' - y), \quad x^y := \prod_{i=1}^d x_i^{y_i}. \quad (1.1)$$

The *stoichiometric subspace* and the *positive stoichiometric class* of $c \in \mathbb{R}_{>0}^d$ are

$$S := \text{span}_{\mathbb{R}}\{y' - y : y \rightarrow y' \in R\}, \quad P := (c + S) \cap \mathbb{R}_{>0}^d.$$

Since $f(x) \in S$, any positive solution starting in P remains in P for as long as it exists. The class P need not be bounded.

A global positive trajectory is *persistent* if $\liminf_{t \rightarrow \infty} x_i(t) > 0$ for every i . These lower bounds may depend on the trajectory. We call the system *permanent on P* if every solution starting in P is global and positive, and there is one compact set $K \subset P$ that eventually contains each such solution. The entry time may depend on the initial point, but K may not. This is the classwise uniformity in the permanence question [2, Definition 4.1].

Theorem 1.1. *Let $C \subset \mathbb{Z}_{\geq 0}^d$ be finite, let $R \subset C \times C$ be weakly reversible, and fix positive rate constants in (1.1). For every positive stoichiometric class P there is a nonempty compact convex set $K_P \subset P$ with the following properties. Every initial point $x^0 \in P$ gives a unique global positive solution, and there is a finite $T(x^0) \geq 0$ such that*

$$x(t) \in K_P \quad (t \geq T(x^0)).$$

Moreover, K_P is forward invariant. In particular, there is $\varepsilon_P \in (0, 1)$, depending only on the network, the fixed rates, and P , such that every such solution satisfies

$$\varepsilon_P \leq x_i(t) \leq \varepsilon_P^{-1} \quad (1 \leq i \leq d, t \geq T(x^0)). \quad (1.2)$$

Thus the fixed-positive-rate, weakly reversible formulation of the permanence conjecture holds, including on unbounded classes. The uniformity in (1.2) is within each fixed class and rate choice; the transient time is allowed to depend on x^0 .

1.2 Historical context and the contribution

The structural theory of mass-action systems begins with the work of Horn and Jackson and of Feinberg [8, 5]. A system is *complex-balanced* if it has a positive state at which the total incoming and outgoing reaction fluxes agree at each complex. For a complex-balanced system, the classical theory gives a unique positive equilibrium in each positive stoichiometric class and an entropy-type Lyapunov function. Complex balance imposes conditions on the rates as well as the graph; weak reversibility by itself does not impose these balance identities. Feinberg's discussion of weakly reversible systems [6, Remark 6.1.E] raised the question of excluding limiting states with a zero concentration. Persistence excludes approach to the boundary even along subsequences, and permanence adds upper bounds and uniformity over initial points.

Important partial results developed both the dynamical picture and the geometric methods. Anderson proved that every positive trajectory of a complex-balanced system with one linkage class converges to its class's positive equilibrium [1, Corollary 4.11]. Craciun, Nazarov, and Pantea proved permanence for two-species weakly reversible systems, as part of their more general theorem for endotactic networks [4, Theorem 6.4]. Their endotactic condition expresses an inward-pointing property of the reactions seen in every direction. Pantea proved persistence of bounded weakly reversible trajectories when $\dim S = 2$, with no restriction to two species [10, Theorem 5.1]. Gopalkrishnan, Miller, and Shiu proved permanence under the geometric condition of strong endotacticity, which includes every weakly reversible network with one linkage class [7, Theorems 1.1 and 1.3]. Boros and Hofbauer later gave another proof of the single-linkage permanence theorem, producing a compact forward-invariant absorbing set [2, Theorem 4.2]. These permanence results also permit rates varying in time within fixed positive bounds.

Craciun's toric-differential-inclusion approach constructs invariant regions by comparing monomials in logarithmic coordinates. The revised manuscript of September 23, 2026 proves persistence for bounded positive trajectories of weakly reversible systems, including rates bounded

above and away from zero; its complex-balanced application establishes global convergence [3, Section 8, final remark]. The general weakly reversible persistence statement there assumes boundedness of the trajectory. Our theorem treats fixed rates and supplies one eventual bound for every trajectory in a class.

The immediate input to the present argument is the affine approximation lemma and trapping construction of [9]. That paper proves that every prescribed positive initial point belongs to a compact convex forward-invariant set contained in the positive orthant. Such sets give boundedness and persistence, but may depend on the initial point. We reproduce the affine lemma and its proof in Section 2, and adapt the trapping construction in Section 3. The additional arguments establish strict increase away from a trapping set and connect trapping sets at different scales. They turn bounds for individual trajectories into permanence on a fixed class. Monomial comparisons and invariant regions have the earlier antecedents just described; the affine approximation lemma is credited to [9, Lemma 2.1 and Corollary 2.2].

1.3 Proof overview

Fix one representative $c \in P$. For $0 < h < 1$, set $B_h = [h, h^{-1}]^d$. Every $x \in B_h$ has a unique representation $x = (h^{p_1}, \dots, h^{p_d})$ with $p \in [-1, 1]^d$. The affine lemma constructs a finite concave minimum

$$F_h(x) = \min_{\lambda \in \Lambda} (r_\lambda \cdot x + b_\lambda(h)),$$

whose slopes $r_\lambda \in [-1, 1]^d$ are independent of h . A label λ is *active* at x when it attains this minimum. For any prescribed tolerance $E : [-1, 1]^d \rightarrow (0, \infty)$, the construction ensures $\|r_\lambda - p\|_\infty < E(r_\lambda)$ for every active label, uniformly for small h . We choose the tolerance so that every nonzero comparison $r_\lambda \cdot v$, where v is a coordinate vector or a difference of distinct complexes, has the same sign as $p \cdot v$. To preserve these signs, our tolerance shrinks near a hyperplane $r \cdot v = 0$; on the hyperplane, the zero comparison imposes no restriction. This is why the affine lemma allows a discontinuous tolerance. When $r_\lambda \cdot v \neq 0$, finiteness of the slope family also gives a uniform positive lower bound for $|p \cdot v|$.

At sufficiently small scales, the labels active at c have zero slope, so $F_h(c)$ is the global maximum of F_h . Its plateau

$$K_h := \{x \in \mathbb{R}^d : F_h(x) = F_h(c)\}$$

is compact and convex, contains c , and satisfies $K_h \subset \text{int } B_{\sqrt{h}}$ (Proposition 3.1). The restriction to P now supplies the decisive strictness condition. A member of the affine family whose slope lies in $S^\perp$ is constant on $c + S$, so its value on P is at least $F_h(c)$. Such a label cannot be active at a point of $P \setminus K_h$.

For any active label λ , order the complexes by the levels $r_\lambda \cdot y$, and separate consecutive levels by cuts. If a downward reaction crosses a cut, its directed return path contains an upward crossing. Monomials at lower source levels dominate those at higher levels, so one upward flux exceeds the entire downward flux for sufficiently small h . The net upward flux across each cut is therefore nonnegative, and is positive whenever a reaction crosses it. Summing these fluxes times the gaps between levels gives $r_\lambda \cdot f(x) \geq 0$, with strict inequality when $r_\lambda \notin S^\perp$. Compactness at each fixed scale makes the strict inequality uniform over all active labels to which it applies (Proposition 3.2). Thus F_h is nondecreasing along solution segments in B_h and increases at a fixed positive rate on $B_h \cap (P \setminus K_h)$. A trajectory in P that stays in B_h for all later times must reach K_h in finite time and remain there.

Every initial point in P belongs to some K_h , giving global existence first. The inclusion $K_h \subset B_{\sqrt{h}}$ then permits passage to a larger scale. Repeated square roots, capped at a scale h_P fixed by the class, reach h_P after finitely many steps. Section 4 carries out this iteration and obtains $K_P = K_{h_P} \cap (c + S)$, with h_P chosen independently of the initial point.

2 Affine minima with arbitrary tolerance

The proof will use a finite minimum of affine functions on the positive orthant. Its essential property is that an active slope approximates the exponent vector of the point where it is active. The allowed error will depend on the slope and may be discontinuous. We prove the construction and its uniform consequence from [9, Lemma 2.1 and Corollary 2.2] in full.

For $m \geq 1$, $u \in \mathbb{R}^m$, and $0 < h < 1$, put

$$X_h(u) = (h^{u_1}, \dots, h^{u_m}).$$

A label $\lambda = (r_\lambda, b_\lambda)$ consists of a slope $r_\lambda \in \mathbb{R}^m$, independent of h , and an offset $b_\lambda(h)$ defined for all sufficiently small positive h . For a finite nonempty family Λ of labels, write

$$F_h(x) = \min_{\lambda \in \Lambda} (r_\lambda \cdot x + b_\lambda(h)).$$

A label is *active at x* if its affine value equals $F_h(x)$. Every minimizing label is active when there is a tie. All offsets in a finite family are understood to be restricted to a common interval on which they are defined.

Lemma 2.1 (Affine approximation with arbitrary tolerance). *Let $m \geq 1$ be an integer, let $a < b$ be real numbers, and let $E : [a, b]^m \rightarrow (0, \infty)$ be any function. There is a finite nonempty family Λ of labels with*

$$r_\lambda \in [a, b]^m, \quad b_\lambda(h) = o(h^a) \quad (h \rightarrow 0+),$$

such that the following implication holds. If $h_\nu \rightarrow 0+$, $p^\nu \in \mathbb{R}^m$ converges to $p \in [a, b]^m$, and a fixed label $\lambda \in \Lambda$ is active at $X_{h_\nu}(p^\nu)$ for every ν , then

$$\|p - r_\lambda\|_\infty < E(r_\lambda).$$

The slopes and the family are independent of h . No regularity is assumed of E , and the approximating vectors p^ν need not lie in $[a, b]^m$.

Proof. We induct on m . Write $\mu(u) = \min_i u_i$ and $\mathbf{1} = (1, \dots, 1)$. Since the largest coordinate of $X_h(u)$ is $h^{\mu(u)}$, we first construct a family for each fixed value t of the smallest exponent. We then extend its guarantee to a neighborhood of t and combine finitely many such families by shifting their offsets. The fixed-minimum construction is a singleton when $m = 1$; the extension and combination therefore also prove the base case.

Throughout the proof, asymptotic comparisons along $h_\nu \rightarrow 0+$ and $p^\nu \rightarrow p$ use fixed gaps between limiting exponents. In particular, if $p_i > q$, then eventually $p_i^\nu \geq q + \delta$ for some $\delta > 0$, so $h_\nu^{p_i^\nu} = o(h_\nu^q)$; if $p_i < q$, the ratio $h_\nu^{p_i^\nu} / h_\nu^q$ tends to infinity. We shall use these comparisons only with strict limiting gaps.

A family for a fixed minimum. Fix $t \in [a, b]$. We construct a finite family Λ_t containing $(t\mathbf{1}, 0)$, with $\mu(r) = t$ for every slope, such that the conclusion of the lemma holds for activity within Λ_t whenever $\mu(p) = t$.

If $m = 1$ or $t = b$, the family $\{(t\mathbf{1}, 0)\}$ suffices: a point $p \in [a, b]^m$ with $\mu(p) = t$ then equals $t\mathbf{1}$. Otherwise $m > 1$ and $t < b$. Say a label has *type* k if exactly k entries of its slope exceed t , the remaining entries being t . Start with the type 0 label $(t\mathbf{1}, 0)$, and construct types $k = 1, \dots, m-1$ successively. After type $k-1$ has been constructed, choose β_k satisfying

$$\begin{aligned} t < \beta_k < b, \quad \beta_k < \beta_{k-1} \quad \text{if } k > 1, \\ \beta_k - t < E(r) \quad \text{for every slope } r \text{ of type } k-1. \end{aligned}$$

These conditions use only finitely many positive values of E , so such a choice is possible. The new type k labels will force every limiting exponent on a coordinate left at t by a type $k-1$ label that stays active to be at most β_k . The last inequality will then bound the error by that label's tolerance.

For each k -element coordinate set $U \subset \{1, \dots, m\}$ and $s \in [\beta_k, b]^U$, let $r(s)$ equal s on U and t off U . Apply the induction hypothesis in dimension k to this cube with tolerance $s \mapsto E(r(s))$, identifying U with its naturally ordered coordinates. For every resulting label (s, d) , add to Λ_t the type k label

$$r = r(s), \quad c(h) = -h^{\beta_k} + d(h), \quad d(h) = o(h^{\beta_k}). \quad (2.1)$$

Here and below c denotes a local offset. The resulting family is finite, and all its offsets are $o(h^t)$. The negative offset favors raising slope entries on coordinates where $X_h(p)$ is small; the decreasing β_k make that preference stronger with each successive type.

To verify the guarantee, let $p^\nu \rightarrow p \in [a, b]^m$ with $\mu(p) = t$, and suppose a type k label (r, c) stays active within Λ_t along $h_\nu \rightarrow 0+$. Let U be its raised coordinate set, and abbreviate $h = h_\nu$. For $k > 0$, comparison with $(t\mathbf{1}, 0)$ gives

$$\sum_{i \in U} (r_i - t) h^{p_i^\nu} \leq h^{\beta_k} - d(h) = (1 + o(1)) h^{\beta_k}.$$

Every coefficient on the left is at least $\beta_k - t > 0$. A limiting exponent $p_i < \beta_k$ would make the corresponding term divided by h^{β_k} tend to infinity. Therefore

$$p_i \geq \beta_k \quad (i \in U). \quad (2.2)$$

Among labels made from this same U , all contributions off U and the term $-h^{\beta_k}$ are common. Thus (s, d) is active in the family furnished by induction. Its limiting exponent vector lies in $[\beta_k, b]^U$, so

$$\max_{i \in U} |p_i - r_i| < E(r).$$

The approximating vectors p_U^ν may approach a face of that cube from outside. This is why the induction statement allows them outside the cube, including when $p_i = \beta_k$ or $p_i = b$.

It remains to bound the coordinates off U . This argument also includes type 0, for which $U = \emptyset$. If $k < m-1$, we claim that

$$p_i \leq \beta_{k+1} \quad (i \notin U).$$

Otherwise choose $j \notin U$ with $p_j > \beta_{k+1}$ and any type $k+1$ label raised on $U \cup \{j\}$. Every limiting exponent on this set exceeds β_{k+1} , by (2.2) and $\beta_k > \beta_{k+1}$ when $k > 0$. The competitor's affine value minus that of $(t\mathbf{1}, 0)$ is therefore

$$-h^{\beta_{k+1}} + o(h^{\beta_{k+1}}).$$

For the supposed active label, the same difference is zero if $k = 0$, and is bounded below by $-h^{\beta_k} + o(h^{\beta_k})$ if $k > 0$. Since $h^{\beta_k} = o(h^{\beta_{k+1}})$, the competitor is strictly smaller for all sufficiently small h along the sequence, a contradiction. Consequently, off U ,

$$|p_i - r_i| \leq \beta_{k+1} - t < E(r).$$

If $k = m - 1$, there is just one coordinate off U . Its limiting exponent is t , since $\mu(p) = t$ and all coordinates on U have limiting exponents greater than t . Its error is zero. This proves the required strict bound for every active label at the fixed minimum.

Extension to nearby minima. For a fixed label $\lambda = (r, c) \in \Lambda_t$, let $A_\lambda \subset [a, b]^m$ consist of the limits of all sequences $p^\nu \in \mathbb{R}^m$ for which $h_\nu \rightarrow 0+$ and λ is active within Λ_t at $X_{h_\nu}(p^\nu)$. This set is closed. Indeed, if $z_n \in A_\lambda$ and $z_n \rightarrow z$, choose from a sequence realizing z_n a parameter $0 < h_n < 1/n$ and a vector u_n with $\|u_n - z_n\|_\infty < 1/n$ such that λ is active at $X_{h_n}(u_n)$. Then $u_n \rightarrow z$, so this selected sequence realizes $z \in A_\lambda$. This argument uses actual active points and requires no continuity of the offsets in h .

The set of limits violating the desired estimate,

$$B_t = \bigcup_{\lambda=(r,c) \in \Lambda_t} \{p \in A_\lambda : \|p - r\|_\infty \geq E(r)\},$$

is compact: the union is finite and each $E(r)$ here is a fixed number. The preceding argument shows that B_t contains no point with $\mu(p) = t$. Continuity of μ and compactness therefore give $\rho_t > 0$ such that B_t contains no point with

$$\mu(p) \in I_t := (t - \rho_t, t + \rho_t).$$

If B_t is empty any positive radius initially suffices; otherwise take $\rho_t < \min_{p \in B_t} |\mu(p) - t|$. Thus Λ_t has the asserted limiting activity property whenever $\mu(p) \in I_t$.

Shrink ρ_t further, if necessary, so that every local offset obeys

$$c(h) = o(h^{t+\rho_t}). \tag{2.3}$$

For a nonsingleton family, choose $t + \rho_t < \min_k \beta_k$ and use (2.1); for a singleton the offset is zero. We have obtained a family valid near each possible minimum exponent. The remaining task is to ensure that a globally active label comes from a family valid at its own limiting point.

Combining the families. Take a finite subcover of $[a, b]$ from the intervals I_t . Discard intervals contained in others, keeping just one if two coincide, and order the remaining centers as $t_1 < \dots < t_N$. Write $I_{t_j} = (L_j, R_j)$. Noncontainment forces both endpoints to have the same strict order; since $t_j = (L_j + R_j)/2$, it is the order of the centers. Hence

$$L_1 < \dots < L_N, \quad R_1 < \dots < R_N.$$

Consecutive intervals overlap. If $R_j \leq L_{j+1}$, a point of $[R_j, L_{j+1}] \subset (t_j, t_{j+1}) \subset [a, b]$ would be uncovered. The first interval contains a and the last contains b .

For $1 \leq j < N$, choose a cut

$$q_j \in (\max\{t_j, L_{j+1}\}, \min\{t_{j+1}, R_j\}).$$

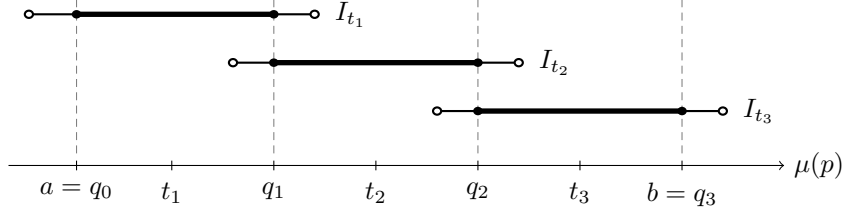


Figure 1: Combining three local families, schematically. Each thick closed interval $[q_{j-1}, q_j]$ lies inside the open neighborhood I_{t_j} where its family is valid. The shifts in (2.5) force the limiting minimum exponent of an active label into this closed interval. At a cut, either neighboring family remains valid.

The interval displayed is nonempty because consecutive intervals overlap and each contains its own center. With $q_0 = a$ and $q_N = b$, we have

$$t_j < q_j < t_{j+1} \quad (1 \leq j < N), \quad [q_{j-1}, q_j] \subset I_{t_j} \quad (1 \leq j \leq N). \quad (2.4)$$

For $N = 1$, the sole interval contains $[a, b]$ and no interior cuts are needed. The closed intervals in (2.4) are the ranges of minimum exponents assigned to the respective families; see Figure 1.

Combine the selected families, replacing every local offset $c(h)$ in Λ_{t_j} by

$$\tilde{c}(h) = c(h) - \sum_{\ell=1}^{j-1} h^{q_\ell}. \quad (2.5)$$

These offsets are $o(h^a)$: (2.3) has $t_j + \rho_{t_j} > a$, and each shift exponent satisfies $q_\ell > t_\ell \geq a$. A common shift preserves activity within each family. Between consecutive families, the new term $-h^{q_j}$ will force any limiting minimum exponent to the correct side of the cut.

Suppose a fixed label from Λ_{t_j} is active in the combined family along $h_\nu \rightarrow 0+$ and $p^\nu \rightarrow p \in [a, b]^m$. Write $c(h)$ for its original local offset and $h = h_\nu$. We claim

$$q_{j-1} \leq \mu(p) \leq q_j.$$

If $j < N$ and $\mu(p) > q_j$, each coordinate h^{p_i} is $o(h^{q_j})$. Also $c(h) = o(h^{q_j})$, by (2.3) and $q_j < R_j = t_j + \rho_{t_j}$. Subtracting the value of the shifted label $(t_{j+1}\mathbf{1}, 0)$ from the active label's value gives

$$(r - t_{j+1}\mathbf{1}) \cdot X_h(p^\nu) + c(h) + h^{q_j} = h^{q_j} + o(h^{q_j}) > 0,$$

a contradiction.

If $j > 1$ and $\mu(p) < q_{j-1}$, compare instead with the shifted label $(t_{j-1}\mathbf{1}, 0)$. The difference is

$$(r - t_{j-1}\mathbf{1}) \cdot X_h(p^\nu) + c(h) - h^{q_{j-1}}.$$

Every coefficient in the scalar product is at least $t_j - t_{j-1} > 0$. Some coordinate has $p_i < q_{j-1}$, so its exponential divided by $h^{q_{j-1}}$ tends to infinity. Meanwhile, $c(h) = o(h^{q_{j-1}})$ because $q_{j-1} < t_j < t_j + \rho_{t_j}$. Thus the displayed difference divided by $h^{q_{j-1}}$ tends to $+\infty$, again contradicting activity. The outer bounds for $j = 1$ or $j = N$ follow from $\mu(p) \in [a, b]$.

By (2.4), the claimed bounds place $\mu(p)$ inside I_{t_j} , including when it equals a cut. The label is active within its original local family, so the guarantee on I_{t_j} gives $\|p - r\|_\infty < E(r)$. This completes the induction. $\square$

The sequential formulation yields the uniform statement needed below. Its order of choices is important: the family, and hence its finite set of slopes, is fixed before the upper bound on h is chosen.

Corollary 2.2 (Uniform activity bound). *Let $m \geq 1$, $a < b$, and $E : [a, b]^m \rightarrow (0, \infty)$ be as in Lemma 2.1. Fix a family Λ supplied by that lemma. There is $h_0 \in (0, 1)$ such that, for every $0 < h < h_0$ and every $p \in [a, b]^m$, each label $\lambda \in \Lambda$ active at $X_h(p)$ satisfies*

$$\|p - r_\lambda\|_\infty < E(r_\lambda).$$

Proof. Otherwise there are $h_\nu \rightarrow 0+$, points $p^\nu \in [a, b]^m$, and active labels λ_ν such that $\|p^\nu - r_{\lambda_\nu}\|_\infty \geq E(r_{\lambda_\nu})$. By finiteness and compactness, pass to a subsequence on which $\lambda_\nu = \lambda$ is fixed and $p^\nu \rightarrow p \in [a, b]^m$. The inequality passes to the limit and contradicts Lemma 2.1. No continuity of E is needed because its argument r_λ is fixed on this subsequence. $\square$

3 Compact plateaux and strict increase

We now apply Corollary 2.2 to the reaction network. For each sufficiently small scale we will construct a compact set in the positive orthant that is forward invariant. For a fixed positive stoichiometric class, we will also prove that a trajectory in that class remaining in the scale's box reaches this set in finite time. The affine minimum and sign argument adapt the trapping construction of [9, Section 3]; the strict increase away from a plateau is the additional property needed for finite entry within a fixed class.

Use X_h from Section 2 with $m = d$, and for $0 < h < 1$ put

$$B_h = [h, h^{-1}]^d.$$

Every $x \in B_h$ is uniquely $X_h(p)$ for $p \in [-1, 1]^d$, with $p = \log x / \log h$, where the logarithm of a vector is taken coordinatewise. We first choose the tolerance in Corollary 2.2 so that active slopes determine both coordinate signs and the ordering of monomials. Let

$$D = \{e_i : 1 \leq i \leq d\} \cup \{z - y : y, z \in C, z \neq y\}, \quad (3.1)$$

where e_i is the i th coordinate vector. This is a finite set of nonzero vectors. We include all pairs of complexes because the flux estimate will compare sources of different reactions, possibly in different components of the reaction graph. Define, for $r \in [-1, 1]^d$,

$$E(r) = \min \left(\left\{ \frac{1}{2} \right\} \cup \left\{ \frac{|r \cdot v|}{2 \|v\|_1} : v \in D, r \cdot v \neq 0 \right\} \right). \quad (3.2)$$

Zero scalar products impose no sign restriction and are omitted. Thus E is positive at every point, including on the comparison hyperplanes; no continuity is needed. Fix the finite nonempty family supplied by Corollary 2.2 for this E , and write its affine functions as

$$L_{\lambda,h}(x) = r_\lambda \cdot x + b_\lambda(h), \quad F_h(x) = \min_{\lambda \in \Lambda} L_{\lambda,h}(x).$$

The slopes $r_\lambda \in [-1, 1]^d$ and the index set Λ are fixed independently of h . A label λ is active at x if $L_{\lambda,h}(x) = F_h(x)$; denote the set of these labels by $A_h(x)$. For every sufficiently small h , the activity estimate is

$$x = X_h(p) \in B_h, \quad \lambda \in A_h(x) \implies \|p - r_\lambda\|_\infty < E(r_\lambda). \quad (3.3)$$

Because the family of slopes is fixed and finite, the number

$$H = \min \left(\{1\} \cup \left\{ \frac{|r_\lambda \cdot v|}{2} : \lambda \in \Lambda, v \in D, r_\lambda \cdot v \neq 0 \right\} \right) \quad (3.4)$$

is positive. If $x = X_h(p) \in B_h$, $\lambda \in A_h(x)$, and $r = r_\lambda$, then

$$|(p - r) \cdot v| \leq \|p - r\|_\infty \|v\|_1 < E(r) \|v\|_1 \leq \frac{|r \cdot v|}{2} \quad (v \in D, r \cdot v \neq 0).$$

Consequently,

$$r \cdot v \neq 0 \implies \operatorname{sgn}(p \cdot v) = \operatorname{sgn}(r \cdot v), \quad |p \cdot v| \geq H \quad (v \in D). \quad (3.5)$$

Fix a positive stoichiometric class P , and choose one representative $c \in P$ once for this class. Choose $h_P \in (0, 1)$ so small that the affine functions are defined and (3.3) holds for every $0 < h \leq h_P$, and also

$$\left\| \frac{\log c}{\log h} \right\|_\infty < H, \quad h^H \sum_{e \in R} k_e < \min_{e \in R} k_e \quad (0 < h \leq h_P). \quad (3.6)$$

The second inequality is imposed only when $R \neq \emptyset$. Such a choice exists because $H > 0$, all the rates are positive, and both $1/|\log h|$ and h^H tend to zero as h tends to zero. In particular, $c \in \operatorname{int} B_h$ for these scales. All the remaining statements of this section concern $0 < h \leq h_P$.

Proposition 3.1 (The compact plateau). *For $0 < h \leq h_P$, let*

$$M_h = F_h(c), \quad K_h = \{x \in \mathbb{R}^d : F_h(x) = M_h\}.$$

There is a constant label with value M_h , and $F_h \leq M_h$ on $\mathbb{R}^d$. The set K_h is nonempty, compact, and convex, and

$$K_h \subset \operatorname{int} B_{\sqrt{h}}. \quad (3.7)$$

Proof. Write $p_c = \log c / \log h$. If a label active at c had a nonzero coordinate $r_{\lambda,i}$, (3.5) with $v = e_i$ would give $|p_{c,i}| \geq H$, contrary to (3.6). Thus every label active at c has zero slope. At least one label is active there, so one of the affine functions is the constant M_h . It follows that $F_h \leq M_h$ everywhere, and that M_h is the least value of any constant label. Hence

$$K_h = \bigcap_{\lambda \in \Lambda} \{x \in \mathbb{R}^d : L_{\lambda,h}(x) \geq M_h\}$$

is closed and convex, and contains c .

If $x = X_h(p) \in K_h \cap B_h$, the constant label with value M_h is active at x . Applying (3.3) to its zero slope gives

$$\|p\|_\infty < E(0) = \frac{1}{2}, \quad \text{and thus} \quad x \in \operatorname{int} B_{\sqrt{h}}.$$

In particular, K_h misses the boundary of B_h . If some point of K_h lay outside B_h , the segment joining it to $c \in \operatorname{int} B_h$ would meet that boundary at a point of K_h , by convexity. This is impossible. Thus $K_h \subset B_h$, and the preceding argument proves (3.7) for every point of K_h . Closedness and boundedness give compactness. $\square$

The plateau is now a compact candidate for a trapping set. A slope in $S^\perp$ gives an affine function that is constant on P . We first exclude such slopes below the plateau, then compare reaction fluxes to prove strict increase for every remaining active label.

Proposition 3.2 (Increase of active affine functions). *Fix $0 < h \leq h_P$. For every $x \in B_h$ and every $\lambda \in A_h(x)$,*

$$r_\lambda \cdot f(x) \geq 0, \quad r_\lambda \cdot f(x) > 0 \quad \text{if } r_\lambda \notin S^\perp. \quad (3.8)$$

Moreover, there is a number $\delta_h > 0$ such that

$$r_\lambda \cdot f(x) \geq \delta_h \quad \text{whenever } x \in B_h, \lambda \in A_h(x), r_\lambda \notin S^\perp. \quad (3.9)$$

At every point of $P \setminus K_h$, all active slopes lie outside $S^\perp$. In particular, (3.9) applies to every active label at each $x \in B_h \cap (P \setminus K_h)$.

Proof. First, let $x \in P \setminus K_h$. If $r_\lambda \in S^\perp$, then $x - c \in S$ implies

$$L_{\lambda,h}(x) = L_{\lambda,h}(c) \geq F_h(c) = M_h.$$

But $F_h(x) < M_h$ by Proposition 3.1. Such a label cannot be active at x .

If R is empty, then $f = 0$ and $S = \{0\}$, so (3.8) holds and (3.9) is vacuous. For the flux comparison, assume $R \neq \emptyset$ and fix $x = X_h(p) \in B_h$ and an active slope r . Whenever $y, z \in C$ satisfy $r \cdot y < r \cdot z$, (3.5) applied to $z - y$ gives

$$\frac{x^z}{x^y} = h^{p \cdot (z-y)} \leq h^H. \quad (3.10)$$

List the distinct values of $r \cdot y$, for $y \in C$, as $a_1 < \dots < a_m$. For $1 \leq j < m$, define the net upward flux across the cut at a_j by

$$Q_j(x) = \sum_{\substack{u \rightarrow u' \in R: \\ r \cdot u \leq a_j < r \cdot u'}} k_{u \rightarrow u'} x^u - \sum_{\substack{u \rightarrow u' \in R: \\ r \cdot u' \leq a_j < r \cdot u}} k_{u \rightarrow u'} x^u.$$

An edge crossing the cut in the first sum is called upward, and an edge in the second sum is called downward.

If there is a downward edge $u \rightarrow u'$, weak reversibility supplies a directed path from u' back to u . This path starts below the cut and ends above it, so it contains an upward edge, say $w \rightarrow w'$. Every source u of a downward edge has $r \cdot u > r \cdot w$. By (3.10) and (3.6),

$$\sum_{\substack{u \rightarrow u' \in R: \\ r \cdot u' \leq a_j < r \cdot u}} k_{u \rightarrow u'} x^u \leq h^H x^w \sum_{e \in R} k_e < k_{w \rightarrow w'} x^w.$$

Thus one upward edge already contributes more than the entire downward sum, and $Q_j(x) > 0$. If there is no downward edge, then $Q_j(x) \geq 0$, with strict inequality whenever there is an upward edge. In all cases $Q_j(x)$ is nonnegative, and it is positive if any reaction crosses the cut.

The contribution of an edge to $r \cdot f(x)$ is its flux multiplied by the difference between the levels of its target and source. Writing this difference as the signed sum of the intervening gaps gives

$$r \cdot f(x) = \sum_{j=1}^{m-1} (a_{j+1} - a_j) Q_j(x). \quad (3.11)$$

Every term is nonnegative. If $r \notin S^\perp$, some reaction $y \rightarrow y'$ satisfies $r \cdot (y' - y) \neq 0$ and therefore crosses at least one cut; the corresponding term is positive. This proves (3.8). When $m = 1$, the sum is empty and every reaction has zero scalar product with r , as required. Self-loops never cross a cut and contribute zero to the vector field.

For the uniform bound, the active set of a fixed label,

$$T_{\lambda,h} = \{x \in B_h : L_{\lambda,h}(x) = F_h(x)\},$$

is compact. If $r_\lambda \notin S^\perp$ and $T_{\lambda,h}$ is nonempty, the continuous function $r_\lambda \cdot f$ is strictly positive on this set and has a positive minimum there. Take the minimum of these positive numbers over the finitely many eligible labels. If there are none, choose any $\delta_h > 0$. This proves (3.9), including the case $R = \emptyset$. $\square$

The strict inequality in Proposition 3.2 is uniform at a fixed scale even near the plateau. Together with the finite nature of the affine minimum, this gives a direct finite-time trapping statement.

Lemma 3.3 (Trapping and finite entry). *Fix $0 < h \leq h_P$. Along any solution segment contained in B_h , the function $t \mapsto F_h(x(t))$ is nondecreasing. If the segment $x([s, t])$ is contained in $B_h \cap (P \setminus K_h)$, then*

$$F_h(x(t)) - F_h(x(s)) \geq \delta_h(t - s). \quad (3.12)$$

Every solution starting in K_h exists for all nonnegative time and remains in K_h . In particular, solutions starting in $K_h \cap P$ remain in $K_h \cap P$. Finally, any global solution in P that stays in B_h from some time onward enters K_h in finite time and then remains there.

Proof. For a solution $x(t)$ and each fixed time t before its maximal existence time, the right derivative of its affine minimum is

$$\frac{d^+}{dt} F_h(x(t)) = \min_{\lambda \in A_h(x(t))} r_\lambda \cdot f(x(t)). \quad (3.13)$$

Indeed, all inactive labels have a positive gap above the minimum at time t . There are finitely many of them, so continuity excludes them from the minimum for a sufficiently short time to the right. For each active label, differentiability of the solution gives

$$L_{\lambda,h}(x(t+s)) = F_h(x(t)) + s r_\lambda \cdot f(x(t)) + o(s) \quad (s \downarrow 0).$$

Taking the minimum of these finitely many expansions proves (3.13).

We use the elementary fact that a continuous function g on $[a, b]$ with right derivative at least q at every point of $[a, b)$ satisfies $g(b) - g(a) \geq q(b - a)$. For completeness, for any $\eta > 0$ the function $g(u) - qu + \eta u$ has positive right derivative before b . Its maximum on the compact interval therefore cannot occur before b . Comparing its values at a and b and then letting $\eta \downarrow 0$ proves the assertion. Formula (3.13) and Proposition 3.2 now give monotonicity in B_h , and give (3.12) when the segment lies in $P \setminus K_h$.

Suppose next that $x(0) \in K_h$, and consider the maximal solution of the polynomial vector field f . Since $K_h \subset \text{int } B_h$, the solution initially stays in B_h . On any interval on which it does so, monotonicity and the global upper bound $F_h \leq M_h$ force $F_h(x(t)) = M_h$. Hence it remains in K_h . It cannot have a first exit from $\text{int } B_h$ within its interval of existence: continuity would place the exit point in both K_h and the boundary of B_h , contradicting Proposition 3.1. Thus the

solution stays in the compact set K_h throughout its maximal interval. A solution of a smooth vector field that remains in a compact subset of its domain extends past any finite endpoint of its interval of existence. To see the relevant point here, f is bounded on a neighborhood of K_h , so a finite endpoint would give a limit of $x(t)$ in K_h , at which local existence continues the solution. The solution is therefore global. It remains positive by (3.7); moreover, $f(x) \in S$ implies $x(t) - x(0) \in S$, so a solution starting in P stays in P .

Finally, let a global solution in P remain in B_h for all $t \geq t_0$. If it never enters K_h after t_0 , then (3.12) gives

$$F_h(x(t)) \geq F_h(x(t_0)) + \delta_h(t - t_0) \quad (t \geq t_0),$$

contradicting $F_h \leq M_h$ for sufficiently large t . It must enter K_h at a finite time, and the forward invariance just proved keeps it there. $\square$

4 A common absorbing set for each class

Fix a positive stoichiometric class P and its representative c , and retain the scale h_P and the functions F_h from Section 3. Lemma 3.3 gives invariance at each scale and finite-time entry whenever a trajectory stays in the corresponding box. Proposition 3.1 connects these two properties through

$$K_h \subset \text{int } B_{\sqrt{h}} \quad (0 < h \leq h_P). \quad (4.1)$$

We now use this relation to replace an initial-point-dependent trapping set by one fixed set for P .

Proof of Theorem 1.1. Let $x^0 \in P$. Since x^0 is positive, we may choose $0 < h_0 \leq h_P$ so small that

$$\left\| \frac{\log x^0}{\log h_0} \right\|_\infty < H,$$

where the logarithm is componentwise and $H > 0$ is the fixed separation constant from Section 3. In particular, $x^0 \in B_{h_0}$. The coordinate sign comparisons force every label active at x^0 to have slope zero, just as they did at c . An active constant label must have the least value among all constant labels, namely $M_{h_0} = F_{h_0}(c)$. Hence

$$x^0 \in K_{h_0}.$$

Lemma 3.3 implies that the solution through x^0 is global, stays positive, and remains in K_{h_0} . Since $f(x) \in S$, it also remains in P .

Starting at h_0 , define

$$h_{j+1} = \min\{\sqrt{h_j}, h_P\}. \quad (4.2)$$

We claim that for every j there is a finite time T_j such that $x(t) \in K_{h_j}$ for all $t \geq T_j$. The case $j = 0$ holds with $T_0 = 0$. Suppose it holds at j . Because $h_{j+1} \leq \sqrt{h_j}$, the boxes satisfy $B_{\sqrt{h_j}} \subset B_{h_{j+1}}$. Thus (4.1) yields

$$x(t) \in B_{h_{j+1}} \quad (t \geq T_j).$$

The finite-entry assertion of Lemma 3.3, followed by its invariance assertion, supplies a finite $T_{j+1} \geq T_j$ with $x(t) \in K_{h_{j+1}}$ for all $t \geq T_{j+1}$. This proves the claim. The argument uses containment in successive boxes; no nesting of the sets K_h is assumed.

Repeated square roots of $h_0 \in (0, 1)$ tend to 1. Since $h_P < 1$, the capped sequence (4.2) therefore reaches h_P after finitely many steps. We conclude that every solution through a point of P enters K_{h_P} in finite time and stays there.

Finally set

$$K_P = K_{h_P} \cap (c + S).$$

This is nonempty because it contains c , and it is compact and convex as the intersection of the compact convex set K_{h_P} with a closed affine subspace. Since K_{h_P} lies in the positive orthant, $K_P \subset P$. Both K_{h_P} and the affine subspace $c + S$ are forward invariant, so K_P is forward invariant. The preceding argument proves its absorbing property. Moreover, $K_P \subset B_{h_P}$, and therefore $\varepsilon_P = h_P \in (0, 1)$ gives

$$\varepsilon_P \leq x_i(t) \leq \varepsilon_P^{-1} \quad (1 \leq i \leq d)$$

for all sufficiently large t , with the entry time allowed to depend on x^0 . The scale h_P , and hence these bounds, was fixed before choosing the initial point. $\square$

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